

assignment10AI

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November 2025

1 Polymarket

In this section, we examine the Polymarket event titled “*What Price Will Bitcoin Hit November 24–30*”. The event metadata and market structure were obtained from the Polymarket Gamma API:

<https://polymarket.com/event/what-price-will-bitcoin-hit-november-24-30/will-bitcoin-reach-9>

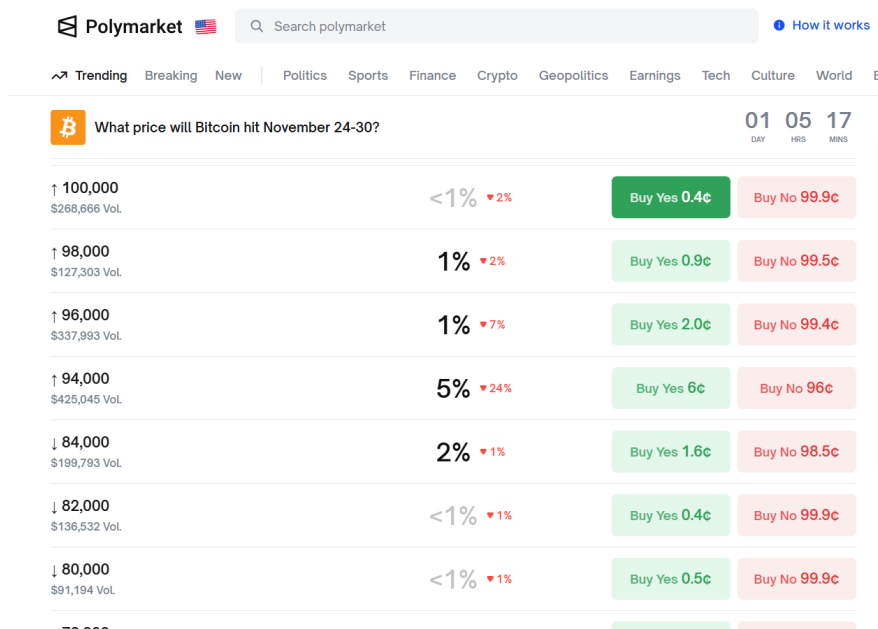


Figure 1: Polymarket order book for the event “What price will Bitcoin hit November 24–30?”

The screenshot shows the active contracts for different Bitcoin price thresholds. Each line lists the implied probability (via the Yes/No prices) and the

trading volume for that contract. Higher thresholds such as \$100,000 trade at very low prices, reflecting the market’s belief that these outcomes are unlikely during the specified window.

1.1 API Query and Returned Markets

To collect market-level data for the event “*What Price Will Bitcoin Hit November 24–30?*”, we queried the Polymarket Gamma API:

<https://gamma-api.polymarket.com/events/slug/what-price-will-bitcoin-hit-november-24-30>

The script extracted all associated markets, parsed the outcomes and prices, and constructed a DataFrame containing each market’s ID, question text, trading volume, the implied Yes price, and the corresponding CLOB token ID. We filtered out incomplete rows and sorted the results by trading volume.

The two highest-volume markets returned by the API were:

- **ID 699721:** “Will Bitcoin reach \$94,000 November 24–30?” This market exhibited the highest trading volume (\$424,436), indicating strong trader engagement. The Yes price was approximately 0.07, implying a market-implied probability of about 7%. The corresponding Yes-side CLOB token ID was retrieved for use in subsequent price-history analysis.
- **ID 699720:** “Will Bitcoin reach \$96,000 November 24–30?” This market recorded the second-highest volume (\$337,746). The Yes price was 0.0135, corresponding to a market-implied probability of roughly 1.35%. Its Yes-side CLOB token ID was also extracted for further analysis.

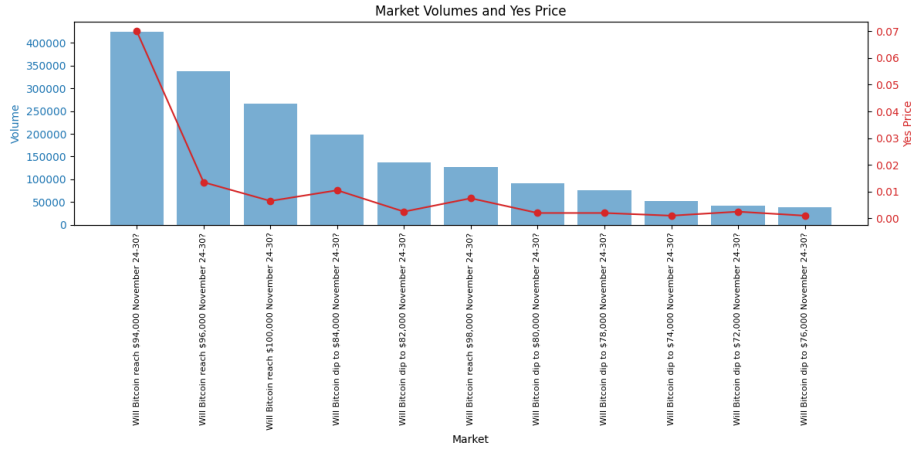


Figure 2: Market volumes (bars) and Yes prices (red line) for all Bitcoin threshold markets in the November 24–30 event.

Figure 2 compares total trading volume with the corresponding Yes prices for each market in the event. The highest-volume markets are those closest to the current Bitcoin price region—for example, the \$94,000 and \$96,000 thresholds—while extreme outcomes, such as a move to \$76,000 or \$100,000, show substantially lower volumes. The Yes prices remain low across all contracts, indicating that the market assigned low probabilities to both large upward and downward moves over the event window.

1.2 Historical Price Data

To examine how market expectations evolved over time, we retrieved historical price data for several selected markets using the Polymarket CLOB price-history endpoint:

https://clob.polymarket.com/prices-history?market={market_id}&startTs=0&interval=max

This API returns the full time series of traded prices for a given market, which we aggregated into hourly medians for clearer visualization. Figure 3 shows the resulting series for four of the most active thresholds: \$94,000, \$96,000, \$100,000, and \$84,000.

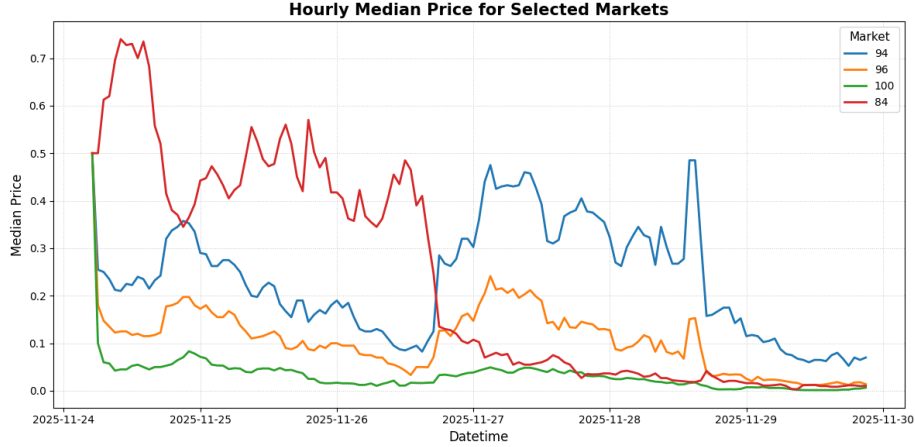


Figure 3: Hourly median prices for selected Bitcoin threshold markets, based on full historical data pulled from the Polymarket CLOB price-history API.

Across the event window, the \$94,000 and \$96,000 markets exhibit the highest and most persistent trading activity, with prices fluctuating between 10% and 45%. In contrast, the \$100,000 and \$84,000 markets decay steadily toward zero as the event progresses, reflecting traders’ decreasing belief in extreme upward or downward moves.

1.3 Market Probabilities and BTC Price Movement

To compare market expectations with actual Bitcoin price movements, we plotted the hourly median probabilities of two key markets—the \$94,000 (Market 94) and \$84,000 (Market 84) thresholds—against the corresponding BTC/USD price series. Figure 4 shows both market probabilities and BTC/USD levels over time.

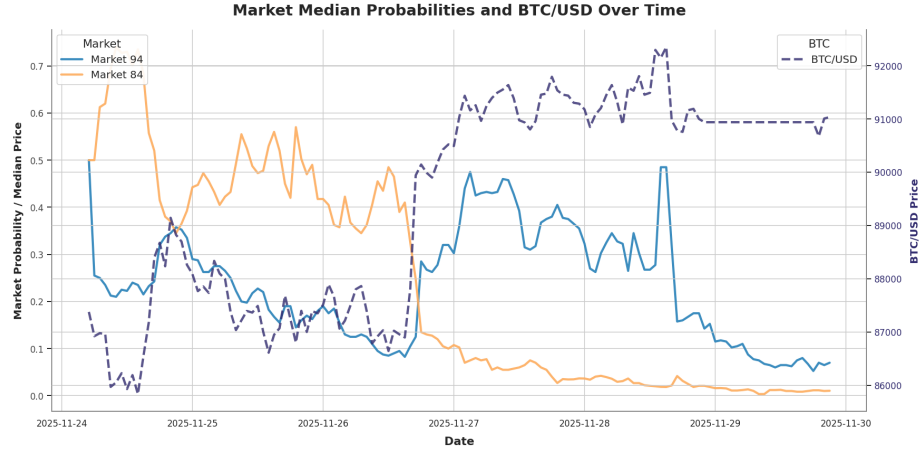


Figure 4: Hourly median probabilities for Market 94 and Market 84 compared with BTC/USD price movements over the November 24–30 event window.

The figure illustrates how changes in Polymarket-implied probabilities tend to move in response to large swings in the underlying BTC price. Market 94 rises when Bitcoin approaches the upper price region, while Market 84 becomes more likely when Bitcoin trades downward.

1.4 Correlation Analysis

To quantify how market probabilities relate to Bitcoin price movements, we computed the pairwise correlations between the median price series of Market 94, Market 84, and the BTC/USD closing price. Figure 5 displays the resulting correlation matrix.

The correlations reveal two clear relationships. Market 94 shows a positive correlation with BTC/USD (0.30), while Market 84 is almost perfectly negatively correlated with BTC/USD (-0.96), reflecting its structure as a downside threshold.

1.5 Lead–Lag Correlation Analysis

To determine whether market probabilities adjust before or after movements in Bitcoin’s price, we compute lead–lag correlations between Market 94, Market 84,

Correlation Matrix: 94, 84, btc_usd_close

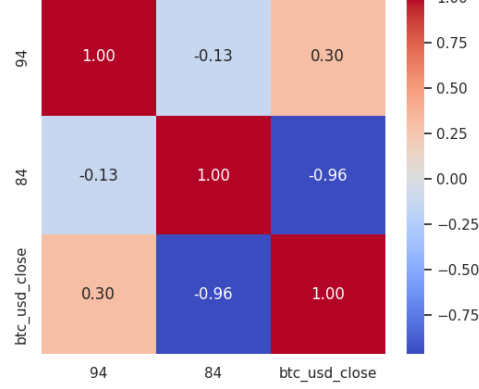


Figure 5: Correlation matrix for Market 94, Market 84, and BTC/USD closing price.

and the BTC/USD closing price. Figure 6 shows the correlations across lags from -8 to $+8$ hours.

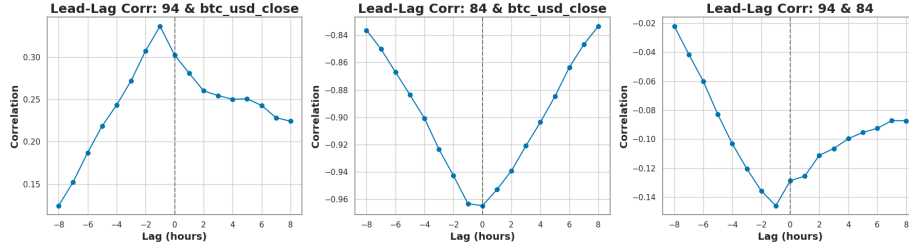


Figure 6: Lead-lag correlations between Market 94, Market 84, and BTC/USD across a ± 8 hour window.

The results show that Market 94 reaches its highest correlation with BTC/USD at a lag of approximately -2 hours, suggesting that increases in the probability of Bitcoin exceeding $\$94,000$ tend to precede upward price movements. Market 84 shows a strong negative lead-lag pattern, indicating that increases in the probability of a dip below $\$84,000$ often lead declines in BTC/USD.

2 Kalshi

2.1 Market 1 — Will SP500 Finish Positive This Year

We analyze the Kalshi binary-market contract:

<https://kalshi.com/markets/kxinxpos/sp500-positive-this-year/kxinxpos-25dec31h1600>

The settlement date is December 31, 2025, at 16:00 (Eastern). At settlement, if the SP 500 closes above its start-of-year value, YES contracts pay \$1.00; otherwise, they pay \$0.00.

2.2 Market Overview: Will the S&P 500 Finish Positive This Year?



Figure 7: Kalshi market page for the contract “Will the S&P 500 finish positive this year?”

Figure 7 shows the Kalshi market interface for the annual S&P 500 outcome contract. The market tracks whether the index will end the year above its January 1st opening level, with the displayed candlestick chart reflecting intraday price movements of the S&P 500. The market price of the YES contract represents the implied probability that the index finishes the year positive, updating continuously as new information affects trader expectations.

2.3 Kalshi API Query and Returned Market

To retrieve information about the active contracts within the S&P 500 positivity series, we queried the Kalshi Trade API using the endpoint:

`https://api.elections.kalshi.com/trade-api/v2/markets?series\_ticker=KXINXPOS&status=open`

This request returns all currently open markets associated with the series KXINXPOS, which corresponds to contracts determining whether the S&P 500 will finish the year above various strike levels. The API output shows that the active market at the time of retrieval was KXINXPOS-25DEC31H1600-T5881.63, with the title “Will the S&P 500 be above 5881.63 on Dec 31, 2025 at 4pm EST?”.

The response also includes the associated event ticker `KXINXP0S-25DEC31H1600`, which groups together all strike-based markets linked to the same year-end S&P 500 outcome.

2.4 Kalshi Market Detail API Response

To obtain detailed information about a specific Kalshi contract, we queried the market-level endpoint:

```
https://api.elections.kalshi.com/trade-api/v2/markets/KXINXP0S-25DEC31H1600-T5881.63
```

The API response provides comprehensive metadata for the binary market “Will the S&P 500 be above 5881.63 on Dec 31, 2025 at 4pm EST?”. Key fields include the contract’s strike level (\$5881.63), last traded price (0.97), and bid-ask spread for both outcomes. The YES side shows a tight market with a bid of 0.96 and an ask of 0.99, implying an approximately 97% market-implied probability that the S&P 500 will finish the year above this threshold. The contract also reports substantial participation, with an open interest of 140,892 shares and liquidity of over \$58,000. Timestamp fields document the market’s creation, opening time, and scheduled expiration, while additional fields such as `can_close_early` and `market_type` confirm this is a standard binary contract with Kalshi’s usual settlement rules.

2.5 Kalshi Candlestick Data API

To analyze the historical evolution of the market price, we queried Kalshi’s candlestick endpoint, which provides aggregated OHLC data (open, high, low, close) over fixed intervals. The request was made to:

```
https://api.elections.kalshi.com/trade-api/v2/series/KXINXY/  
markets/{event_ticker}/candlesticks
```

Using daily intervals (1440 minutes) and a specified start date, the API returns a time series of candlesticks covering the full trading history for the selected event. Each entry includes the day’s opening price, high and low trade, and closing price, along with the corresponding timestamp. This allows us to track how the market-implied probability has evolved since the contract’s listing.

Figure 8 plots the daily close prices for the market “Will the S&P 500 be above 5881.63 on Dec 31, 2025?”. The trend shows a steady increase in the contract’s value from around 0.40 in early May to above 0.95 by late November, indicating that traders have become increasingly confident that the S&P 500 will finish the year above the specified threshold.

2.6 Kalshi Volume Over Time

In addition to price data, the candlestick endpoint also provides information on daily trading volume, allowing us to analyze how participation in the market



Figure 8: Daily close prices for the Kalshi contract, based on candlestick data returned by the Kalshi API.

changed over time. Using the same API call to the Kalshi candlestick endpoint, the “volume” field was extracted for each daily interval and plotted in Figure 9.

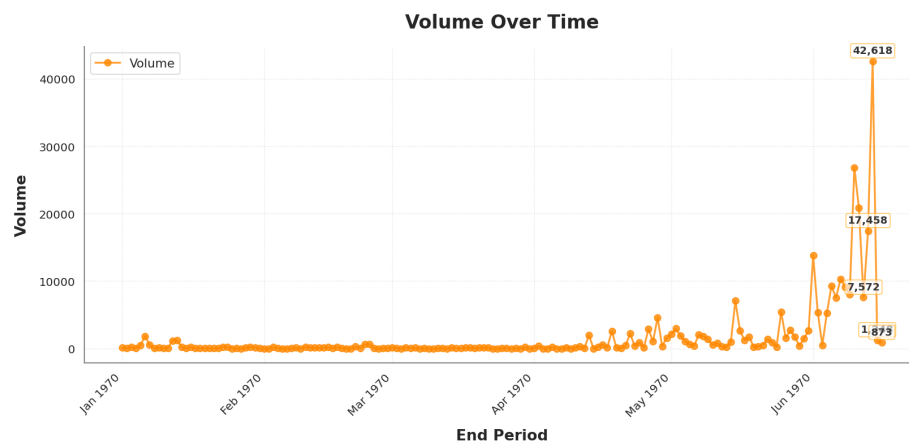


Figure 9: Daily trading volume for the Kalshi market “Will the S&P 500 be above 5881.63 on Dec 31, 2025?”.

The volume series shows minimal early trading activity, with volumes generally below 500 contracts during the first several months. Participation increased substantially in late May and June, with spikes above 10,000 contracts and a notable peak exceeding 40,000. This pattern suggests that liquidity and trader engagement rose significantly as the year progressed and the S&P 500 continued to strengthen, increasing interest in the year-end positivity contract.

3 Underlying Asset Analysis

To understand how the Kalshi market tracks its underlying asset, we compare the normalised daily close price of the contract “Will the S&P 500 be above 5881.63 on Dec 31, 2025?” with the normalised S&P 500 index itself. Normalisation rescales both series to lie between 0 and 1, allowing their movements to be compared directly despite being on different price scales.

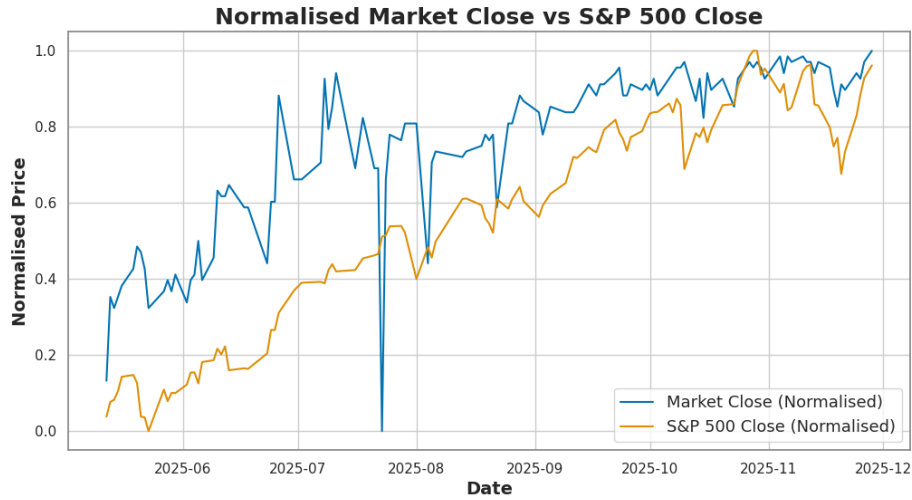


Figure 10: Normalised contract close price versus normalised S&P 500 index.

The figure shows that the Kalshi market closely co-moves with the S&P 500, especially from mid-year onward. As the index trends upward through the summer and into the autumn, the probability of finishing above the strike increases accordingly, with the contract price rising toward 1. Occasional divergences reflect market-specific dynamics, such as liquidity variation or short-term sentiment shifts, but overall the contract behaves as a monotonic transformation of the underlying index path. This confirms that the binary market efficiently incorporates movements in the underlying asset into its implied probability.

3.1 Underlying Asset Volume Comparison

To examine how trading activity in the Kalshi market relates to activity in the underlying equity market, we compare the normalised daily trading volume of the contract with the normalised volume of SPY, the most heavily traded S&P 500 ETF. Normalising both series enables a direct comparison of relative fluctuations in trading intensity over time.

The comparison shows that SPY exhibits consistently high and relatively stable volume throughout the period, reflecting its status as a major market instrument. In contrast, Kalshi trading volume remains low for much of the

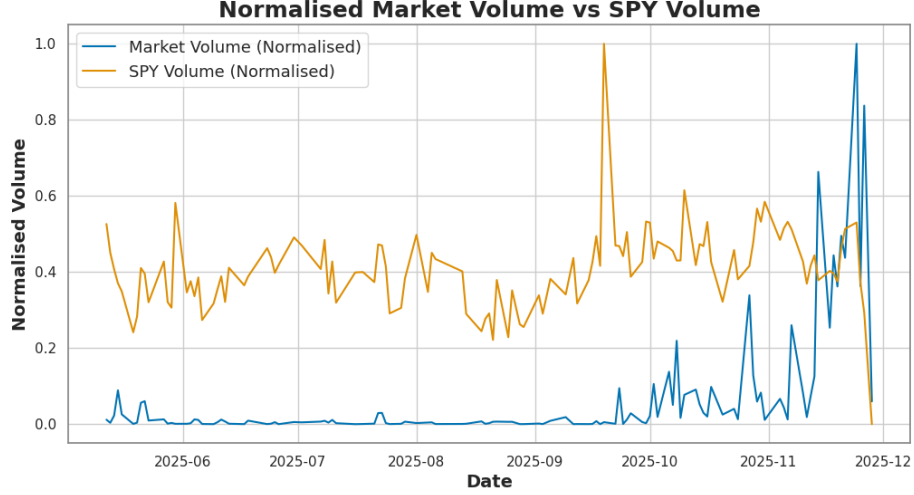


Figure 11: Normalised Kalshi market volume versus normalised SPY trading volume.

year before rising sharply in the final months as the event date approaches and the S&P 500 trends closer to the strike level. This late-period surge in Kalshi volume indicates growing participant interest and increased conviction as more information about year-end market performance becomes available.

3.2 Correlation Between Market Prices, S&P 500, and Volume

To quantify the relationships among the Kalshi market, the underlying S&P 500 index, and trading activity, we compute the pairwise correlations between the normalised contract close price, the S&P 500 close, the Kalshi market volume, and SPY trading volume. The resulting correlation matrix is shown in Figure 12.

The contract close price is strongly correlated with the S&P 500 close (0.87), indicating that the market price closely tracks movements in the underlying index. Both volume measures show weaker correlations with price variables, reflecting the fact that trading activity is influenced by event proximity and trader participation rather than day-to-day index fluctuations. The low correlation between Kalshi market volume and SPY volume (0.077) suggests that activity in the prediction market is driven by its own dynamics rather than general equity market turnover.

3.3 Rolling Close–Volume Relationship

To examine how trading activity co-moves with changes in market pricing, we compute a 14-day rolling correlation between the Kalshi market’s daily close

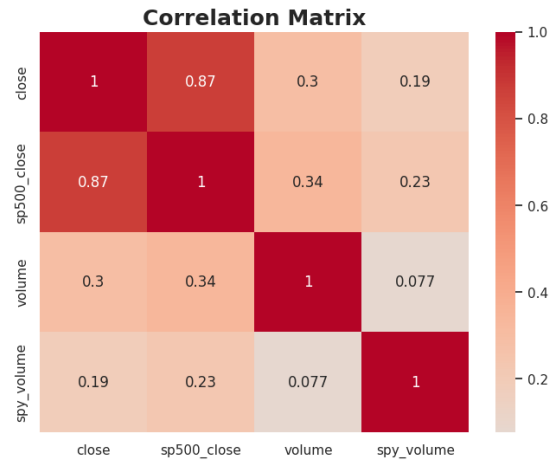


Figure 12: Correlation matrix comparing Kalshi market close price, S&P 500 close, Kalshi volume, and SPY volume.

price and its corresponding trading volume. This measure captures whether periods of increasing price are typically associated with rising participation, or whether volume tends to emerge during sell-offs. Figure 13 shows the resulting rolling correlation series.

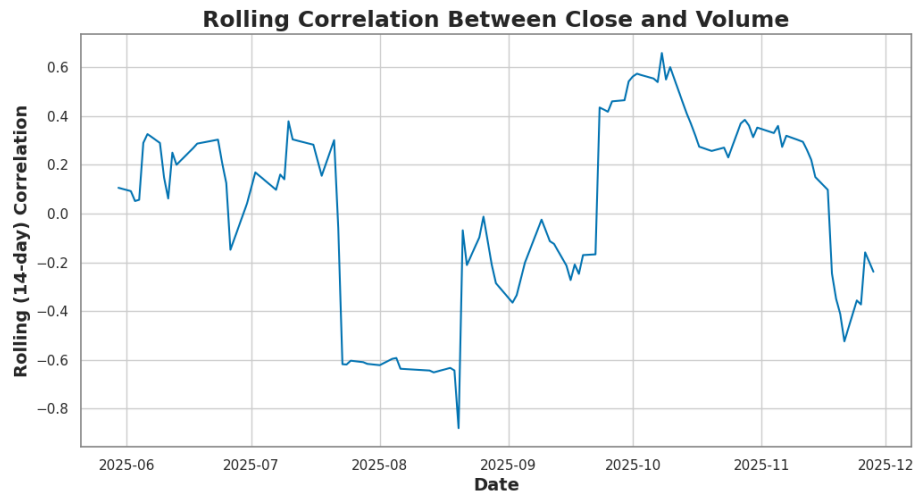


Figure 13: 14-day rolling correlation between Kalshi market close price and trading volume.

The rolling correlation fluctuates substantially across the observation window. During early and mid-summer, the correlation is weakly positive, indi-

cating that higher prices are modestly associated with higher trading activity. Entering late July and August, however, the correlation turns sharply negative, suggesting that large bursts of volume tended to occur during price declines—possibly reflecting event-related uncertainty or position unwinding.

In the autumn months, the correlation reverses again and becomes strongly positive, reaching values above 0.6. This shift implies that traders increasingly entered the market during upward price movements, consistent with rising confidence that the S&P 500 would finish the year positive. Toward late November, the correlation weakens once more, illustrating that price–volume dynamics in prediction markets are not stable but vary depending on market conditions, trader sentiment, and event proximity.

3.4 Predictive Feature Correlations

To evaluate which variables are most useful for forecasting the Kalshi market’s implied daily close probability, we constructed a set of predictive features—including lagged prices, volume measures, returns, moving averages, spreads, and interaction terms—and computed their pairwise correlations. Figure 14 (14.png) summarizes these relationships. The matrix reveals strong clustering among Kalshi close prices, S&P 500 close levels, and their lagged and smoothed variants, reflecting the tight coupling between the market’s payoff structure and movements in the underlying index. In contrast, volume-based features such as SPY volume and Kalshi market volume show weaker correlations with price-driven predictors, suggesting they contribute independent information related to liquidity and trading activity. Interaction and spread features fall between these extremes, exhibiting moderate correlations that indicate potential value in capturing nonlinear behavior. Overall, the feature set displays a balance between highly persistent price dynamics and lower-correlation volume components, making it well-suited for modeling daily shifts in implied probability.

3.5 In-Sample Prediction of Implied Probability

To assess how well a simple linear regression model can explain movements in the Kalshi market’s daily implied probability, we fit the model using all engineered features and compared the predicted close values to the actual observed closes. Figure 15 (15.png) shows the resulting in-sample time series. The predicted series tracks the actual probabilities closely, capturing the gradual upward drift in the market as well as most short-term fluctuations. Although the model smooths out some of the more abrupt dips—reflecting the limitations of a linear specification in handling sudden regime shifts—it nonetheless reproduces overall dynamics with high fidelity, indicating that a substantial portion of daily variation in the implied probability can be explained by lagged market data and movements in the S&P 500.

The linear regression model provides insight into which factors most strongly influence the daily implied probability of the Kalshi S&P 500 market. The co-

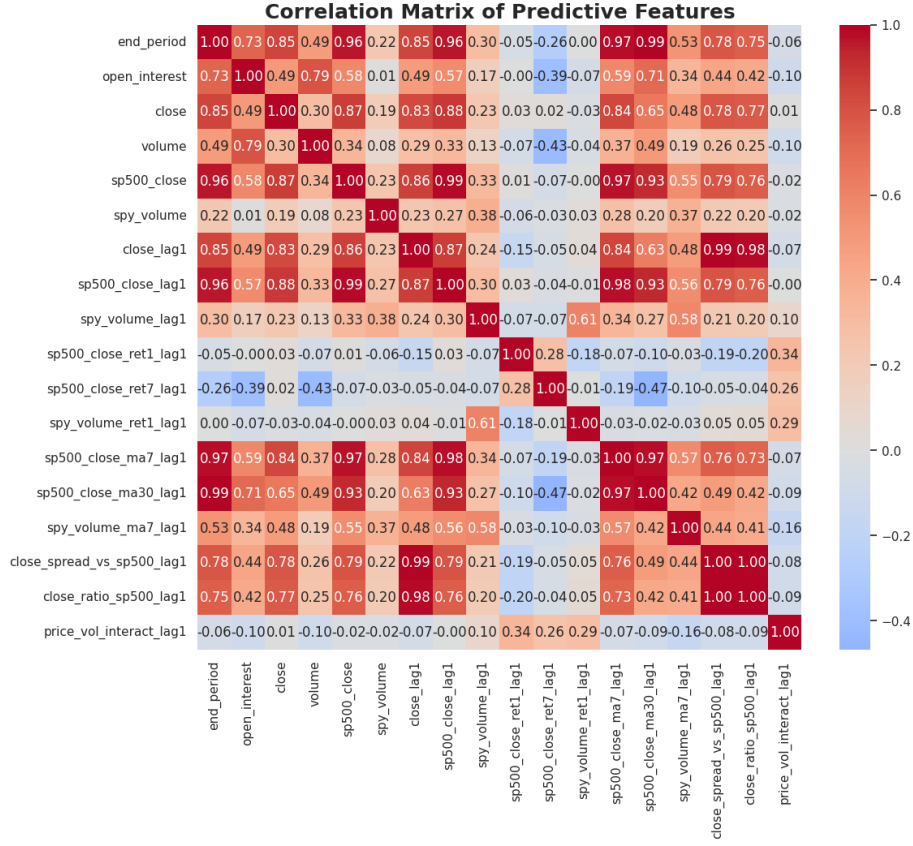


Figure 14: Correlation matrix of engineered predictive features for modeling the Kalshi implied probability.

efficients indicate that past market behavior plays a dominant role: both the lagged close and the lagged close–spread relative to the S&P carry large negative weights (approximately -0.93), while the ratio of the Kalshi close to the S&P close has a strong positive effect (1.30), suggesting that relative mispricing between the market and its underlying index is an important predictor. Short-term index momentum is also influential, with `sp500_close_ret1_lag1` contributing a sizable positive coefficient (0.97), indicating that upward movement in the S&P tends to push the Kalshi contract toward “Yes.” Interaction effects between price and volume matter as well, reflected in the large positive coefficient on `price_vol_interact_lag1` (5.81). Other features, such as lagged SPY volume or long-horizon moving averages, contribute minimally. Overall model fit is moderate, with $R^2 = 0.5137$ showing that roughly half of the variation in implied probability is explained by the predictors, and an RMSE of 0.0657 indicating reasonably tight in-sample accuracy.

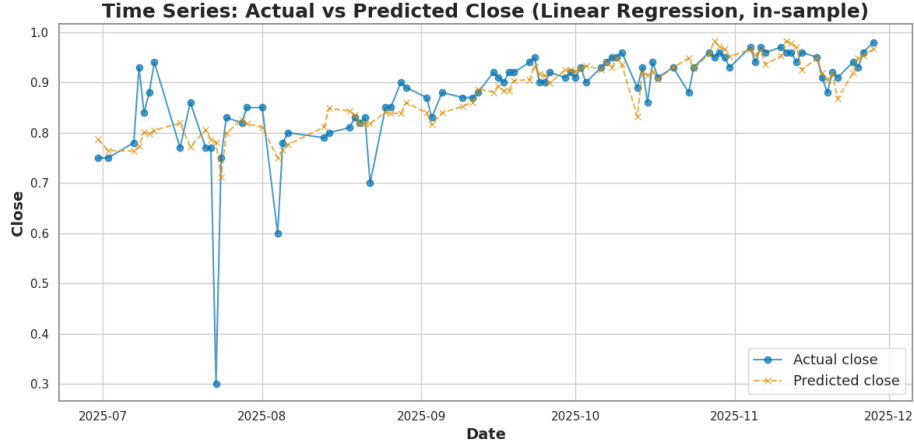


Figure 15: In-sample comparison between actual and predicted Kalshi market close probabilities using a linear regression model.

4 Federal Reserve Decision Markets

To examine expectations around upcoming Federal Reserve policy decisions, we queried all active markets in the `KXFEDDECISION` series using the Kalshi API endpoint:

`https://api.elections.kalshi.com/trade-api/v2/markets?series\_ticker=KXFEDDECISION&status=open`

The response returns every currently tradable FOMC contract across meetings from December 2025 into 2027, each defined by an event ticker and a specific rate-change outcome (e.g., cut by 25bps, hike by 0bps, etc.). For each contract, we extracted its name, trading volume, and displayed liquidity.

Overall, the December 2025 meeting shows by far the most active trading, with several markets exceeding four to six million contracts in volume. These include “Hike 0bps,” “Cut 25bps,” and “Cut 50bps,” reflecting concentrated trader interest in the near-term decision. Markets for later meetings (January 2026 through late 2027) exhibit extremely high posted liquidity but very low or zero volume, indicating that while institutional liquidity providers are active, actual trading activity remains sparse until those meetings approach. In summary, the API output reveals a maturity structure typical of prediction markets: heavy activity in the nearest events, and thin participation further out on the calendar.

4.1 December 2025 FOMC Markets

Within the `KXFEDDECISION` series, the December 2025 meeting shows the highest concentration of trading activity. Five mutually exclusive outcomes are listed

for this event, covering all standard rate-move possibilities. The API output for these markets is:

- **KXFEDDECISION-25DEC-C26:** Cut rates by >25bps Volume: 4,447,259
Liquidity: 42,663,805
- **KXFEDDECISION-25DEC-C25:** Cut rates by 25bps Volume: 5,689,991
Liquidity: 22,117,255
- **KXFEDDECISION-25DEC-H0:** Hike rates by 0bps (no change) Volume: 6,548,879 Liquidity: 33,604,391
- **KXFEDDECISION-25DEC-H25:** Hike rates by 25bps Volume: 183,867
Liquidity: 109,542,260
- **KXFEDDECISION-25DEC-H26:** Hike rates by >25bps Volume: 44,374
Liquidity: 109,818,979

Trading activity is heavily concentrated in the “Cut” and “No-change” markets, each with several million contracts traded, indicating strong trader engagement and active price discovery around these policy scenarios. The “Hike” markets, while displaying very high posted liquidity, show much lower actual volume, suggesting that rate hikes are viewed as less likely for this meeting. Overall, the December 2025 FOMC markets are the deepest and most informative in the series, reflecting their proximity and relevance to current macroeconomic expectations.

4.2 Close-Price Dynamics for December 2025 Outcomes

To study how the market’s expectations evolved over time, we retrieved the historical close prices for all five December 2025 FOMC outcome markets (cut >25bps, cut 25bps, no-change, hike >25bps, and hike 25bps). The resulting time series are shown in Figure 16.

Figure 16 reveals a clear ordering of scenarios in the market’s perceived likelihood. The 25bps cut emerges as the dominant outcome throughout the sample, with its price rising from roughly 60% in early September to more than 85% by late October before experiencing a temporary dip in mid-November. The “no-change” (0bps) contract fluctuates between 15% and 50%, reflecting shifting macroeconomic narratives regarding inflation persistence and recession risk. Larger cuts (>25bps) show a steady decline in probability as economic data increasingly suggested a more measured policy path. Both hike outcomes remain priced effectively at zero throughout, consistent with strong trader consensus that tightening was off the table. Overall, the combined time series captures how new information—ranging from inflation prints to employment data and Federal Reserve communications—continuously reshaped the pricing of plausible policy moves for the December 2025 meeting.

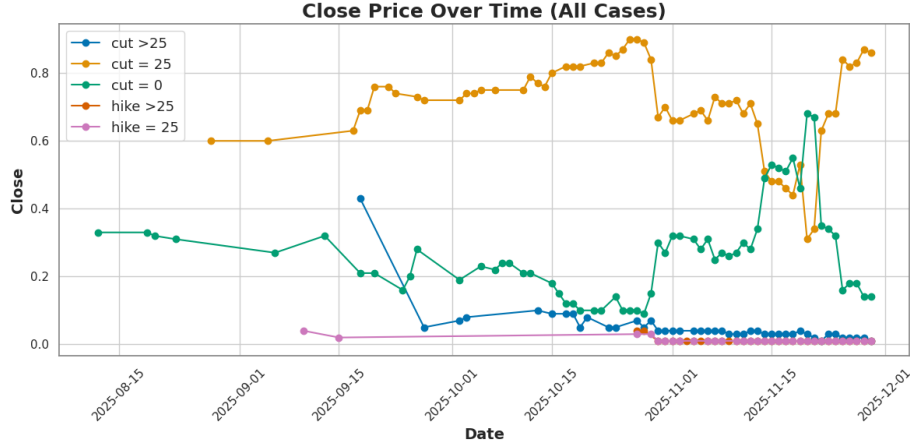


Figure 16: Close prices over time for all December 2025 FOMC rate-move outcomes.

4.3 Volume Dynamics for December 2025 Outcomes

Trading activity across the five December 2025 FOMC markets exhibits a sharply nonlinear pattern, with volumes remaining negligible through early autumn before accelerating dramatically in the weeks prior to the meeting (Figure 17). The surge begins in late October and peaks in mid-November, a period coinciding with major macro releases (CPI, PCE, payrolls) and intensified Fed communications. The no-change (0bps) and 25bps cut contracts dominate participation, each reaching single-day volumes above 500,000 contracts, highlighting traders' focus on the central policy fork. Larger cuts (>25bps) attract substantial but more volatile bursts of trading, consistent with hedging behavior or opportunistic positioning. Both hike markets again show almost no sustained activity, reflecting near-unanimous expectations that tightening was off the table. Overall, the temporal clustering of volume emphasizes how market participants concentrate liquidity around pivotal data moments and narrative shifts as the meeting approaches.

4.4 CME FedWatch Implied Probabilities

To contextualize Kalshi pricing, Figure 18 overlays the CME FedWatch Tool's implied probabilities for the December 2025 target-rate ranges of 350–375 bps, 375–400 bps, and 400–425 bps. Throughout early 2025, the distribution remains relatively balanced across all three outcomes, reflecting uncertainty around the pace of disinflation. By mid-year, however, the 375–400 bps range steadily strengthens as the modal path, consistent with market expectations for modest easing but not an aggressive cutting cycle. In late August a sharp repricing occurs: the probability of 350–375 bps (a full quarter-point deeper cut) surges

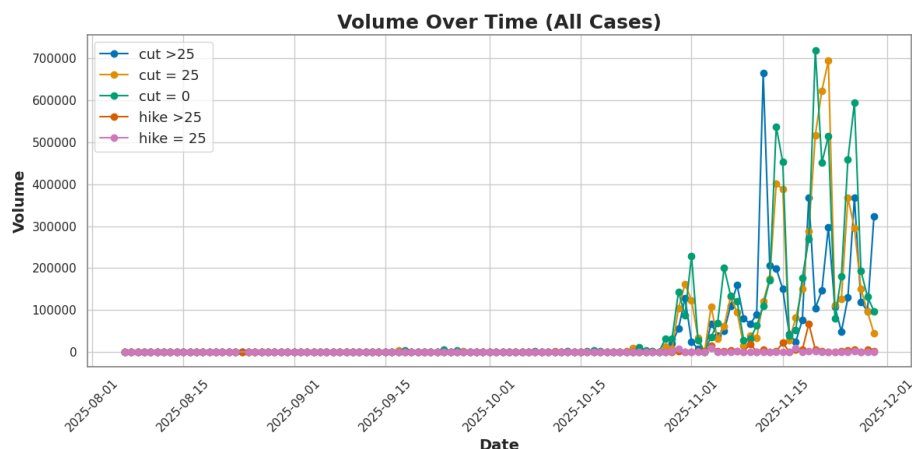


Figure 17: Trading volume over time for all December 2025 FOMC rate-move outcomes.

substantially, mirroring both softer macro releases and a dovish shift in rate-path commentary. From September onward, FedWatch probabilities increasingly flatten the upper tail, with the 400–425 bps scenario collapsing toward zero as traders effectively rule out further tightening. The CME curve therefore provides an external benchmark that largely corroborates Kalshi’s implied odds, especially the growing conviction toward modest or moderate cuts into year-end.

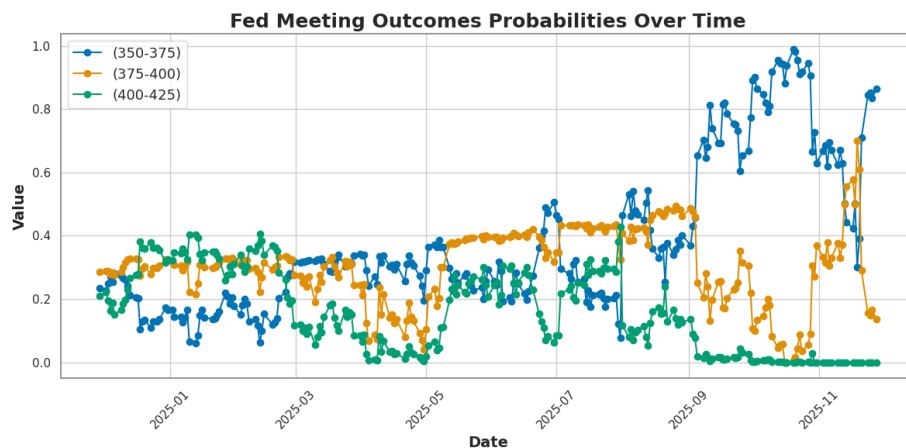


Figure 18: CME FedWatch implied probabilities for December 2025 target-rate outcomes.

4.5 Kalshi vs. CME FedWatch (Post-Oct 2025)

Figure 19 overlays Kalshi’s market-implied closing probabilities for each December 2025 rate outcome with the corresponding CME FedWatch estimates. Beginning in early October, Kalshi pricing strongly favors a 25 bp cut, with the “cut = 25” contract trading persistently above 0.75, while FedWatch assigns a more diffuse distribution across 350–375 and 375–400 bps. As macro conditions soften into late October, both sources converge: Kalshi’s pricing drifts lower for the “cut > 25” and “cut = 0” outcomes, while FedWatch sharply increases the probability of the 350–375 bps range. Through November, divergences tighten further—Kalshi continues to assign high probability to a single dominant outcome, whereas FedWatch reflects more uncertainty via a broader two-way distribution. Overall, the joint series illustrate that Kalshi tends to move earlier and more decisively, while FedWatch adjusts more gradually, offering a complementary benchmark for rate expectations.

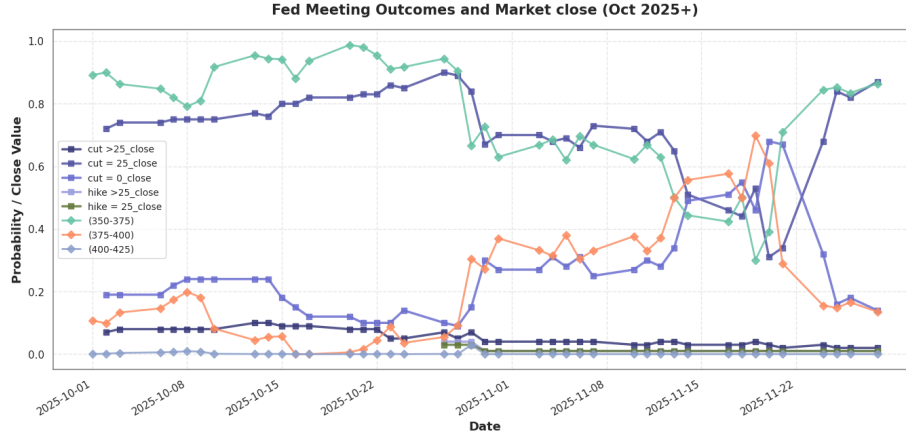


Figure 19: Kalshi market closes vs. CME FedWatch implied probabilities (Oct 2025 onward).

4.6 Correlation Between Kalshi Outcomes and CME FedWatch Bins

Figure 20 presents the correlation matrix between the Kalshi December 2025 outcome contracts and the CME FedWatch implied probabilities for the corresponding rate ranges. Several clear structures emerge. First, the Kalshi “cut = 25” and “cut = 0” contracts exhibit a nearly perfect negative correlation ($\rho \approx -0.99$), reflecting the mutually exclusive nature of adjacent rate outcomes within the same meeting. Similarly, the FedWatch (350–375) and (375–400) bins are almost perfectly negatively correlated, indicating that shifts in the market’s perceived modal outcome occur through weight transfers between neighboring

ranges. Second, Kalshi’s hiking contracts (“hike >25” and “hike = 25”) are nearly perfectly correlated with each other ($\rho = 1.0$), suggesting that markets treat any hiking scenario as a unified low-probability cluster rather than differentiating between magnitudes. Finally, cross-platform relationships reveal strong alignment: for instance, “cut = 25” correlates positively with the FedWatch (375–400) bin and negatively with (350–375), mirroring the directional shift in expected terminal rates. Overall, the structure demonstrates that Kalshi and FedWatch encode the same macro state transitions, but Kalshi expresses them more discretely through binary contracts.

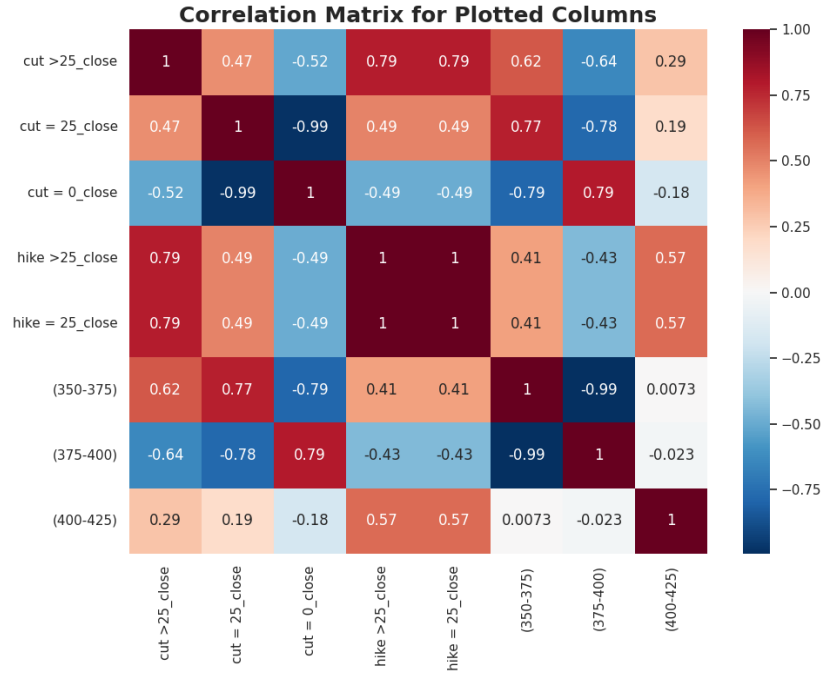


Figure 20: Correlation matrix between Kalshi December 2025 outcomes and CME FedWatch rate-range probabilities.